

CU Swap — Project Report

Team Number: 011-3

Team Name: CU Swap

Course: CSCI 3308 — Fall 2025

Team Members

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Title

CU Swap — A Marketplace for CU Boulder Students

Project Description

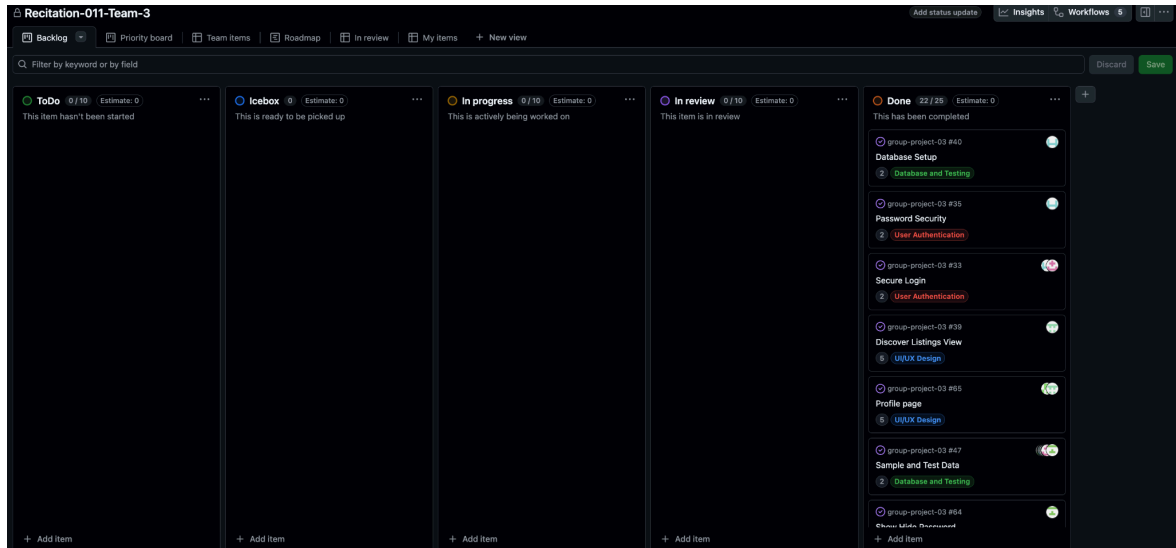
CU Swap is a campus-focused marketplace for University of Colorado Boulder students to buy and sell items safely within the CU community. Users create accounts with their @colorado.edu emails, verify ownership via a one-time code, and then post listings with photos, prices, categories, and descriptions. Buyers can browse by category, view seller ratings, and view contact information of the sellers to arrange on-campus meetups for exchanges and/or questions..

We believe the app's value is that its student-only scope listings stay relevant to campus life (textbooks, dorm gear, lab equipment, etc), with faster discovery and simpler coordination than large public marketplaces. To maintain trust and safety, CU Swap includes moderator accounts with tools to remove inappropriate content and restrict abusive users.

CU Swap uses Node.js/Express, Handlebars for server-side views, PostgreSQL for persistence, Stripe for demo payments, and session-based authentication. We followed Agile practices with short sprints, a project board for tasks, peer reviews via pull requests, and user acceptance tests mapped to core user stories (register, list, browse, purchase, review, and moderate). The result is a working, deployed application built for the CU Boulder community.

Project Tracker — GitHub Project Board

- Link to Project Board: <https://github.com/orgs/CU-CSCI3308-Fall2025/projects/26>
- Screenshot:



Demo Video

Side note for the demo video – our recording software for some reason didn't capture any dropdown menus from our application, however they are there.

- Video Link:
https://drive.google.com/file/d/1NmmOR8UZPTS3E7FM0CXDI959IzfnRxBP/view?usp=drive_link

Version Control (VCS)

- Repository Link: <https://github.com/CU-CSCI3308-Fall2025/group-project-03>
 - Repo contains: source code (/src), test cases (UAT artifacts), demo video, README.md, documentation, and the project board link.
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Contributions

Sashi Wolf

Implemented the my reviews page where a user can view and delete their reviews that they've written. Implemented the my listings page where a user can view, edit, and delete their listings that they've posted. Implemented the listings page when a user clicks on a listing, and connected that to the stripe api that deals with payment and connected it to the leave reviews page.

Hayden Leovy

Implemented login backend, helped with log in register front end backend integration, leave review frontend backend implemented, create listing frontend backend review, created categories dropdown menu.

Shon Butz

Implemented the front-end of the home page, the login page, and the registration page. Did the email verification through nodemailer. Implemented admin controls and admin rendering based on variables. Redid some css style in some pages.

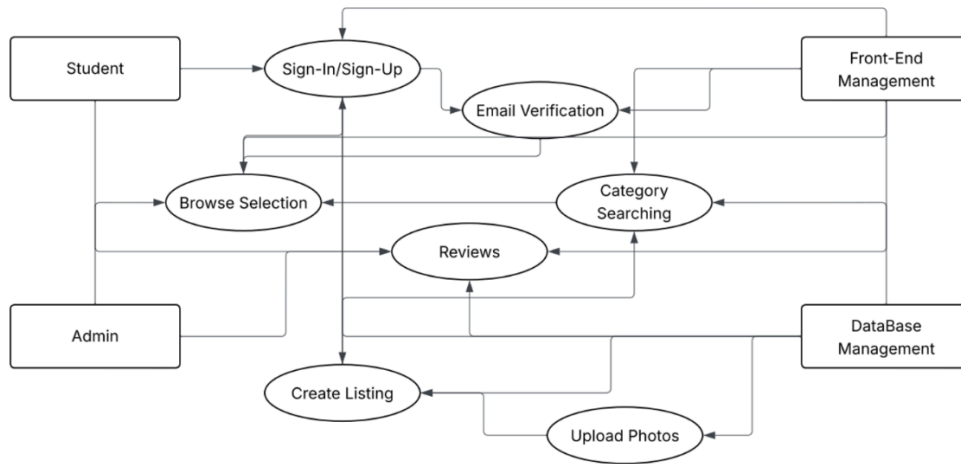
Dylon Gregory

Implemented the checkout page through stripe with error handling for payment failure and miss information. Implemented the category sort filter on the discover page and the category for the create listings page.

Joe Young

Implemented the project's discovery page and the project's profile page. Implemented the discovery page to show up to 100 items in the database, including sales item thumbnails, names, and prices. Implemented the profile page to display user information and implemented the UI for updating profile information.

Use Case Diagram



Wireframes

Home page:

Register Login

CU swap
Info about website

Create listing:

Nav bar goes here

Create listing

Title
Description
Price
category
URL

Submit

Discover:

Nav bar goes here

Category
Dropdown

Create listing



Title
price



Title
price



Title
price

Edit listing:

Nav bar goes here

Edit listing

Title

Description

URL

Price

category

Cancel

Save

Edit profile:

Nav bar goes here

Username

Email

Phone

Current password

New Password

Confirm new password

Cancel

Save

Leave review:

Nav bar goes here

Star rating

Text review

Leave review

Login:

Nav bar goes here

Login

Email

Password

Register here

Logout:

Nav bar goes here

Logged out successfully

Login again

Mod listings:

Nav bar goes here

Category Dropdown

Create listing



Title
price
Delete



Title
price
Delete



Title
price
Delete

Mod users:

Nav bar goes here

Registered users

Username
Phone
Email
Delete

Username
Phone
Email
Delete

My listings:

Nav bar goes here

My listings



Title
Description
Category
View Edit Delete listing



Title
Description
Category
View Edit

My Reviews:

Nav bar goes here

My reviews

Reviewed user
Delete review
Star rating
Delete review

Reviewed user
Star rating
Delete review

Payment:

Nav bar goes here

Checkout

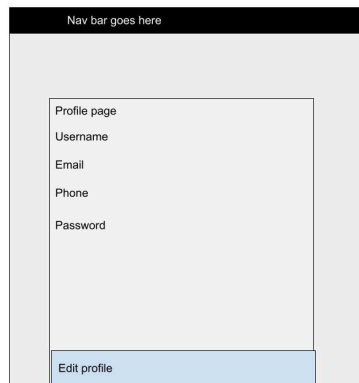
Item details

Item cost

Card details

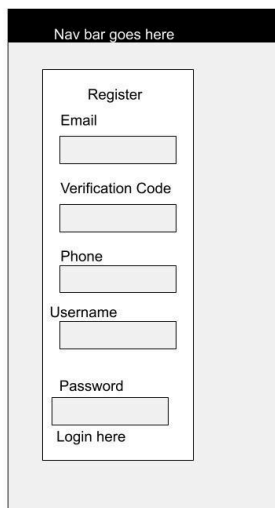
Click here to pay

Profile:



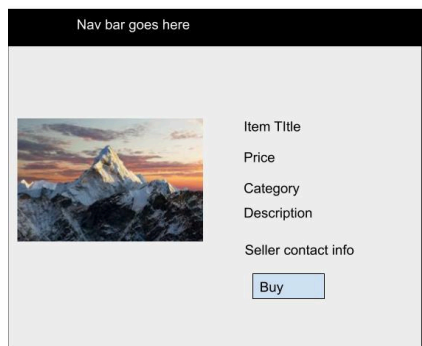
A diagram of a profile page layout. It features a black header bar at the top with the text "Nav bar goes here". Below the header is a light gray container. Inside this container is a white box labeled "Profile page". Within the "Profile page" box, the labels "Username", "Email", "Phone", and "Password" are listed vertically. At the bottom of the "Profile page" box is a blue button labeled "Edit profile".

Register:



A diagram of a register form layout. It features a black header bar at the top with the text "Nav bar goes here". Below the header is a light gray container. Inside this container is a white box labeled "Register". Within the "Register" box, the labels "Email", "Verification Code", "Phone", "Username", and "Password" are listed vertically, each followed by a text input field. At the bottom of the "Register" box is a link labeled "Login here".

View Listing:



A diagram of a view listing layout. It features a black header bar at the top with the text "Nav bar goes here". Below the header is a light gray container. Inside this container, on the left, is a placeholder image of a mountain. To the right of the image, the labels "Item Title", "Price", "Category", "Description", and "Seller contact info" are listed vertically. At the bottom of the listing is a blue button labeled "Buy".

User Acceptance Testing — Results & Observations

User Observation Report

During the User Acceptance Testing (UAT) process, structured observations were conducted to evaluate how users interacted with the application and to determine whether their actions

aligned with the intended system behavior. The following report summarizes user actions, motivations, consistency with defined use cases, and any resulting application changes.

1. User Actions

Throughout the testing sessions, users performed all tasks outlined in the UAT scenarios. These tasks included registering new accounts, logging in with provided credentials, navigating the system interface, completing purchases using Stripe test card information, and leaving or deleting reviews. Each action was carried out in accordance with the instructions defined in the test plan.

2. User Reasoning

User actions were guided by the goals and requirements of each test scenario. During purchase-related tasks, users completed transactions for the purpose of accessing the “Leave a Review” feature. In the registration feature tests, users intentionally entered both valid and invalid data to verify the correctness of system validation, error handling, and success messaging. All user reasoning aligned with the intent of the corresponding test cases.

3. Consistency With Use Cases

User behavior remained consistently aligned with the expected use cases. Participants interacted with the interface intuitively, demonstrating no difficulty in navigating between pages or initiating the required workflows. Their interactions matched the system flows described in the feature requirements, confirming that the application’s user experience supports its intended functionality.

4. Deviations From Expected Actions

No deviations from expected actions were observed. The interface design closely resembles common e-commerce platforms, which allows users to instinctively understand how to complete tasks without confusion. As a result, participants followed the anticipated workflows without needing additional clarification or instruction.

5. Resulting Application Changes

No modifications to the application were required following UAT. All features performed as expected, and the behaviors observed during testing aligned with the projected outcomes listed in the Acceptance Test Plan. Because no deviations, errors, or usability issues occurred, the system was deemed stable and ready for deployment.

Deployment

Hosted App: <https://cu-swap.onrender.com/>